DO NOW · AO1 REVISION

Nuclear fission and fusion

Atomic structure
Paper 1 · Separate Physics only
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Define nuclear fission.**	The splitting of a large, unstable nucleus into two smaller nuclei.	☐
2	**Give two examples of elements that can undergo nuclear fission.**	Uranium and plutonium	☐
3	**What is spontaneous fission?**	Fission that happens without the nucleus absorbing a neutron.	☐
4 ★	**State what usually has to happen before a nucleus undergoes fission.**	The unstable nucleus must absorb a neutron.	☐
5 ★	**State what a nucleus splits into during fission.**	Two smaller nuclei, two or three neutrons and gamma rays	☐
6 ★	**What is a chain reaction?**	Neutrons released by one fission cause further fissions, which release more neutrons, and so on.	☐
7 ★	**Define nuclear fusion.**	The joining of two light nuclei to form a heavier nucleus.	☐
8	**In nuclear fusion, which has more mass: the two light nuclei or the nucleus they form?**	The two light nuclei	☐
9 ★	**Some mass is lost during nuclear fusion. State what it is converted into.**	Energy (of radiation)	☐
10	**Name the energy store that all fission products have.**	The kinetic energy store	☐
11 ★	**State how the chain reaction in a reactor differs from that in a nuclear weapon.**	Reactor: controlled. Weapon: uncontrolled, causing an explosion.	☐
12	**Name the process that releases energy in stars.**	Nuclear fusion	☐

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